

# Readmission Patterns in ICU Patients

A SQL Analysis of MIMIC-III Clinical Data

## Q1 Top 20 Most Readmitted Patients

### Description

Finds the 20 patients with the most hospital admissions. Groups admissions by patient and filters to those with more than one visit.

### SQL Query

```
SELECT
  subject_id,
  COUNT(hadm_id) AS num_admissions
FROM admissions
GROUP BY subject_id
HAVING COUNT(hadm_id) > 1
ORDER BY num_admissions DESC
LIMIT 20
```

| Patient ID | Admissions |
|------------|------------|
| 13033      | 42         |
| 109        | 34         |
| 11861      | 34         |
| 5060       | 31         |
| 20643      | 24         |
| 19213      | 23         |
| 7809       | 22         |
| 5727       | 21         |
| 23657      | 20         |
| 11318      | 19         |
| 23707      | 17         |
| 73713      | 17         |
| 19059      | 17         |
| 7275       | 16         |
| 4787       | 16         |
| 41976      | 15         |
| 10635      | 14         |
| 3100       | 14         |
| 25225      | 14         |
| 19620      | 14         |

### Interpretation

The most-admitted patient returned 42 times; the bottom of the top 20 had 14. A small group of patients accounts for a disproportionate share of hospital visits.

## Q2

## Distribution of Time Between Admissions

## Description

Measures how many days passed between each patient's discharge and their next admission. Buckets the gaps to show whether returns happen quickly or over time.

## SQL Query

```
WITH gaps AS (  
  SELECT subject_id,  
         DATE_DIFF(DATE(admittime),  
                   DATE(LAG(disctime) OVER  
                     (PARTITION BY subject_id  
                      ORDER BY admittime)), DAY)  
         AS days_since_last_discharge  
  FROM admissions  
)  
SELECT gap_bucket,  
       COUNT(*) AS num_admissions,  
       ROUND(COUNT(*) / SUM(COUNT(*)  
        OVER()) * 100, 1) AS pct  
FROM gaps g JOIN repeat_patients r  
  ON g.subject_id = r.subject_id  
WHERE days_since_last_discharge >= 0  
GROUP BY gap_bucket  
ORDER BY gap_bucket
```

| Time Since Last Discharge | Admissions | % of Readmissions |
|---------------------------|------------|-------------------|
| Within 7 days             | 1424       | 11.4              |
| 8 to 30 days              | 1995       | 16.0              |
| 31 to 90 days             | 2197       | 17.6              |
| 91 to 180 days            | 1468       | 11.8              |
| 181 to 365 days           | 1535       | 12.3              |
| Over 365 days             | 3835       | 30.8              |

## Interpretation

27% of returns happen within 30 days — rapid bounce-backs. But the biggest group (31%) waited over a year, suggesting many readmissions reflect long-term chronic disease rather than discharge failures.

## 30-Day Readmission Rate by Admission Type

### Description

Calculates what percentage of patients in each admission category returned within 30 days of discharge.

### SQL Query

```
WITH gaps AS (  
  SELECT subject_id, hadm_id,  
         admission_type,  
         DATE_DIFF(DATE(admittime),  
                   DATE(LAG(dischtime) OVER  
                     (PARTITION BY subject_id  
                      ORDER BY admittime)), DAY)  
         AS days_gap  
  FROM admissions  
)  
SELECT admission_type,  
       COUNT(*) AS total_admissions,  
       SUM(CASE WHEN days_gap <= 30  
              THEN 1 ELSE 0 END) AS readmits,  
       ROUND(SUM(CASE WHEN days_gap <= 30  
                  THEN 1 ELSE 0 END) / COUNT(*) * 100, 1)  
       AS readmit_rate_pct  
FROM gaps  
GROUP BY admission_type  
ORDER BY readmit_rate_pct DESC
```

| Admission Type | Total Admissions | 30-Day Readmits | Readmit Rate % |
|----------------|------------------|-----------------|----------------|
| EMERGENCY      | 42071            | 3149            | 7.5            |
| URGENT         | 1336             | 73              | 5.5            |
| ELECTIVE       | 7706             | 167             | 2.2            |
| NEWBORN        | 7863             | 32              | 0.4            |

### Interpretation

Emergency patients return at the highest rate (7.5%) — over 3x the elective rate (2.2%). Nearly 3,150 emergency patients came back within a month.

## Description

Identifies the top 10 diagnoses among patients who returned within 30 days. Joins diagnosis codes to their readable labels.

## SQL Query

```
WITH gaps AS (
  SELECT subject_id, hadm_id,
         DATE_DIFF(DATE(admittime),
                   DATE(LAG(dischtime) OVER
                     (PARTITION BY subject_id
                      ORDER BY admittime)), DAY)
         AS days_gap
  FROM admissions
)
SELECT d.short_title,
       COUNT(*) AS readmit_count
FROM gaps g
JOIN diagnoses_icd dx
  ON g.hadm_id = dx.hadm_id
  AND dx.seq_num = 1
JOIN d_icd_diagnoses d
  ON dx.icd9_code = d.icd9_code
WHERE g.days_gap <= 30
GROUP BY d.short_title
ORDER BY readmit_count DESC
LIMIT 10
```

| Primary Diagnosis        | Readmissions |
|--------------------------|--------------|
| Septicemia NOS           | 210          |
| Acute respiratry failure | 130          |
| Other postop infection   | 109          |
| Fetal/neonatal jaund NOS | 105          |
| Food/vomit pneumonitis   | 87           |
| CHF NOS                  | 61           |
| Gastrointest hemorr NOS  | 57           |
| Acute kidney failure NOS | 55           |
| Pneumonia, organism NOS  | 52           |
| Acute & chronc resp fail | 51           |

## Interpretation

Sepsis, respiratory failure, and postoperative infection top the list — all conditions with chronic disease roots. These aren't one-off events; they reflect underlying conditions that weren't fully resolved at discharge.

## Q5

## Gender Breakdown: Readmitted vs Single-Admission

## Description

Compares how many male vs female patients were readmitted versus admitted only once. Uses an outer join so all patients are counted.

## SQL Query

```
SELECT p.gender,
       CASE
         WHEN a.num_adm > 1
           THEN 'Readmitted'
         WHEN a.num_adm = 1
           THEN 'Single Admission'
         ELSE 'No Admission Record'
       END AS readmit_status,
       COUNT(*) AS patient_count
FROM patients p
LEFT JOIN (
  SELECT subject_id,
         COUNT(hadm_id) AS num_adm
  FROM admissions
  GROUP BY subject_id
) a ON p.subject_id = a.subject_id
GROUP BY p.gender, readmit_status
ORDER BY p.gender, readmit_status
```

| Gender | Status           | Patients | % of Gender |
|--------|------------------|----------|-------------|
| F      | Readmitted       | 3347     | 16.4        |
| F      | Single Admission | 17052    | 83.6        |
| M      | Readmitted       | 4190     | 16.0        |
| M      | Single Admission | 21931    | 84.0        |

## Interpretation

Male and female patients are readmitted at nearly the same rate (16.0% vs 16.4%). Gender is not a useful predictor — admission type matters far more.

## Q6

## Lab Test Repetition Frequency Within One Admission

## Description

Counts how many times the same lab test was ordered in a single admission. Groups by frequency range to show how common heavy repetition is.

```
WITH test_counts AS (
  SELECT l.subject_id, l.hadm_id,
         TRIM(UPPER(d.label)) AS test_name,
         COUNT(*) AS times_tested
  FROM labevents l
  JOIN d_labitems d
    ON l.itemid = d.itemid
  WHERE l.hadm_id IS NOT NULL
  GROUP BY
    l.subject_id, l.hadm_id, test_name
)
SELECT
  CASE
    WHEN times_tested >= 100 THEN '100+'
    WHEN times_tested >= 50  THEN '50-99'
    WHEN times_tested >= 20  THEN '20-49'
    WHEN times_tested >= 10  THEN '10-19'
    ELSE '6-9'
  END AS repetition_bucket,
  COUNT(*) AS num_pairs,
  ROUND(COUNT(*) / SUM(COUNT(*))
        OVER()*100, 1) AS pct
  FROM test_counts WHERE times_tested > 5
GROUP BY repetition_bucket
```

| Draws per Test per Admission | Patient-Test Pairs | % of Total |
|------------------------------|--------------------|------------|
| 100+ draws                   | 6408               | 0.6        |
| 50 to 99 draws               | 34428              | 3.2        |
| 20 to 49 draws               | 193610             | 18.1       |
| 10 to 19 draws               | 391261             | 36.6       |
| 6 to 9 draws                 | 442739             | 41.4       |

## Interpretation

Most repeated tests fall in the routine range (6–19 draws). However, 6,408 admission-test combinations had the same test drawn 100+ times in one stay — driven by critically ill patients needing hourly blood gas monitoring.

## Q7

## Avg Lab Draws by Readmission Frequency Group

## Description

Compares lifetime lab draw totals across patients grouped by how many times they were admitted. Shows whether heavier testing comes from more visits or more intensive care per visit.

```
WITH admission_counts AS (
  SELECT subject_id,
    COUNT(hadm_id) AS num_admissions
  FROM admissions GROUP BY subject_id
),
lab_counts AS (
  SELECT subject_id,
    COUNT(*) AS total_lab_draws
  FROM labevents
  WHERE hadm_id IS NOT NULL
  GROUP BY subject_id
)
SELECT
  CASE
    WHEN a.num_admissions=1
      THEN 'Single Admission'
    WHEN a.num_admissions<=3
      THEN '2-3 Admissions'
    WHEN a.num_admissions<=6
      THEN '4-6 Admissions'
    WHEN a.num_admissions<=10
      THEN '7-10 Admissions'
    ELSE '10+ Admissions'
  END AS readmission_group,
  COUNT(*) AS num_patients,
  ROUND(AVG(l.total_lab_draws),0)
  AS avg_total_labs
FROM admission_counts a
JOIN lab_counts l
  ON a.subject_id = l.subject_id
```

| Admission Group    | Patients | Avg Total Labs | Avg Labs/Admission |
|--------------------|----------|----------------|--------------------|
| Single Admission   | 38665    | 348.0          | 348.0              |
| 2 to 3 Admissions  | 6501     | 966.0          | 437.0              |
| 4 to 6 Admissions  | 867      | 2150.0         | 478.0              |
| 7 to 10 Admissions | 122      | 3289.0         | 410.0              |
| 10+ Admissions     | 46       | 5049.0         | 347.0              |

## Interpretation

Patients with 10+ admissions accumulate 14.5x more lifetime lab draws than single-admission patients. Notably, the number of labs per individual stay stays roughly the same across all groups — the extra burden comes entirely from returning more often.



## Q8 Recurring Abnormal Glucose: Population Distribution

### Description

Looks at patients with 10 or more glucose tests on record and calculates what percentage of those tests came back abnormal. Groups patients into ranges from 0% to 100% abnormal.

```
WITH glucose_summary AS (  
  SELECT l.subject_id,  
         COUNT(*) AS total_draws,  
         SUM(CASE WHEN l.flag='abnormal'  
           THEN 1 ELSE 0 END) AS abnml,  
         ROUND(SUM(CASE WHEN l.flag=  
           'abnormal' THEN 1 ELSE 0 END)  
           /COUNT(*)*100,1) AS abnml_pct  
  FROM labevents l  
  JOIN d_labitems d  
    ON l.itemid=d.itemid  
  WHERE d.label LIKE '%Glucose%'  
    AND l.hadm_id IS NOT NULL  
  GROUP BY l.subject_id  
  HAVING COUNT(*) >= 10  
)  
SELECT  
  CASE WHEN abnml_pct=100  
    THEN '100% Abnormal' ... END  
  AS abnormal_bucket,  
  COUNT(*) AS patient_count,  
  ROUND(COUNT(*)/SUM(COUNT(*)  
    OVER()*100,1) AS pct_of_patients  
FROM glucose_summary  
GROUP BY abnormal_bucket
```

| Abnormal Range  | Patients | % of Total | Avg Draws | Avg Abnormal % |
|-----------------|----------|------------|-----------|----------------|
| 100% Abnormal   | 667      | 2.7        | 14.4      | 100.0          |
| 75-99% Abnormal | 9532     | 37.9       | 30.3      | 83.8           |
| 50-74% Abnormal | 10558    | 42.0       | 34.8      | 63.4           |
| 25-49% Abnormal | 3678     | 14.6       | 29.0      | 38.8           |
| 1-24% Abnormal  | 689      | 2.7        | 21.2      | 17.1           |
| 0% Abnormal     | 35       | 0.1        | 12.2      | 0.0            |

### Interpretation

80% of frequently-tested patients had abnormal glucose results more than half the time. Only 35 patients out of 25,000+ were never abnormal. Most patients leave the hospital with glucose that still isn't under control.

## Q9

## Same Primary Diagnosis on Consecutive Admissions

## Description

Compares each patient's primary diagnosis to their previous admission's diagnosis. Flags cases where the exact same condition brought them back.

```
WITH primary_dx AS (
  SELECT a.subject_id, a.hadm_id,
  a.admittime, dx.icd9_code,
  d.short_title
  FROM admissions a
  JOIN diagnoses_icd dx
    ON a.hadm_id=dx.hadm_id
    AND dx.seq_num=1
  JOIN d_icd_diagnoses d
    ON dx.icd9_code=d.icd9_code
),
with_lag AS (
  SELECT subject_id, hadm_id,
  icd9_code, short_title,
  LAG(icd9_code) OVER
  (PARTITION BY subject_id
  ORDER BY admittime)
  AS prev_icd9_code
  FROM primary_dx
)
SELECT recurrence_flag,
  COUNT(*) AS num_admissions,
  ROUND(COUNT(*)/SUM(COUNT(*))
  OVER()*100,1) AS pct
FROM with_lag
WHERE prev_icd9_code IS NOT NULL
GROUP BY recurrence_flag
```

| Recurrence          | Admissions | % of Total |
|---------------------|------------|------------|
| Same Diagnosis      | 1566       | 13.0       |
| Different Diagnosis | 10517      | 87.0       |

## Interpretation

13% of readmissions — 1,566 cases — returned with the identical primary diagnosis. These are patients whose core condition wasn't resolved between visits, representing the most directly preventable category of readmissions.

## Q10 High-Utilizer Overlap: Frequent Readmissions + Heavy Lab Testing

### Description

Finds patients who appear in both the frequent-readmission group and the heavy lab-testing group. Ranks them by a combined score of admissions and lab draws.

```
WITH frequent_readmitters AS (  
  SELECT subject_id,  
         COUNT(hadm_id) AS num_admissions  
  FROM admissions  
  GROUP BY subject_id  
  HAVING COUNT(hadm_id) > 2  
)  
frequent_testers AS (  
  SELECT subject_id,  
         COUNT(*) AS num_lab_draws  
  FROM labevents  
  WHERE hadm_id IS NOT NULL  
  GROUP BY subject_id  
  HAVING COUNT(*) > 200  
)  
SELECT r.subject_id,  
       r.num_admissions,  
       t.num_lab_draws,  
       RANK() OVER (  
         ORDER BY r.num_admissions DESC,  
                  t.num_lab_draws DESC)  
       AS utilizer_rank  
FROM frequent_readmitters r  
JOIN frequent_testers t  
  ON r.subject_id = t.subject_id  
ORDER BY utilizer_rank  
LIMIT 20
```

| Rank | Patient ID | Admissions | Lab Draws |
|------|------------|------------|-----------|
| 1    | 13033      | 42         | 7872      |
| 2    | 109        | 34         | 7483      |
| 3    | 11861      | 34         | 7208      |
| 4    | 5060       | 31         | 3466      |
| 5    | 20643      | 24         | 4375      |
| 6    | 19213      | 23         | 9653      |
| 7    | 7809       | 22         | 6373      |
| 8    | 5727       | 21         | 4810      |
| 9    | 23657      | 20         | 3362      |
| 10   | 11318      | 19         | 10080     |
| 11   | 73713      | 17         | 5119      |
| 12   | 19059      | 17         | 3697      |
| 13   | 23707      | 17         | 2567      |
| 14   | 4787       | 16         | 4458      |
| 15   | 7275       | 16         | 2292      |
| 16   | 41976      | 15         | 4131      |
| 17   | 25225      | 14         | 10892     |
| 18   | 19620      | 14         | 5777      |
| 19   | 3100       | 14         | 4168      |
| 20   | 10635      | 14         | 3110      |

The highest-burden patients show up in both lists. Patient 13033 had 42 admissions and 7,872 lab draws — the clearest example of a high-utilizer whose repeated returns drove compounding resource consumption across their entire hospital history.

# Readmission Patterns in ICU Patients: A Cohesive Story

Q1

## Who comes back?

Top 20 patients: up to 42 admissions each

Q2

## How fast?

27% return within 30 days;  
31% after 1+ year

Q3

## Which pathway?

Emergency: 7.5% 30-day rate vs Elective: 2.2%

Q4

## Why?

Sepsis, resp failure & postop infection lead

Q5

## Who?

Gender neutral: M 16.0% vs F 16.4%

Q6

## Testing burden

0.6% of test pairs: 100+ draws in one stay

Q7

## Lifetime labs

10+ admissions → 14.5x more lifetime labs

Q8

## Unresolved?

80% of patients: glucose abnormal >50% of draws

Q9

## Same problem?

13% return with identical primary diagnosis

Q10

## Who is highest risk?

Overlap: most admissions + most lab draws

*A small subset of patients with chronic, poorly-controlled conditions cycles repeatedly through emergency admissions — and the data to identify them proactively exists in the EHR.*